

Problem

A sinusoidal source supplies 100 kVAR reactive power to load $\mathbf{Z} = 250\angle -75^\circ \Omega$.

Determine: (a) the power factor, (b) the apparent power delivered to the load, and (c) the rms voltage.

Step-by-step solution

Step 1 of 3

(a)

Calculate the power factor:

$$\begin{aligned}\text{pf} &= \cos(-75^\circ) \\ &= 0.2588 \text{ leading}\end{aligned}$$

The power factor is leading because the current leads the voltage.

Therefore, the power factor is $\boxed{0.2588 \text{ leading}}$.

Step 2 of 3

(b)

Calculate the apparent power S .

$$\begin{aligned}S &= \frac{Q}{\sin \theta} \\ &= \frac{100 \text{ kVAR}}{\sin(75^\circ)} \\ &= \frac{100 \times 10^3}{0.9659} \\ &= 103.53 \text{ kVA}\end{aligned}$$

Therefore, the value of apparent power S is $\boxed{103.53 \text{ kVA}}$.

Step 3 of 3

(c)

Calculate the value of rms voltage.

$$\begin{aligned}S &= |\mathbf{V}_{\text{rms}}| |\mathbf{I}_{\text{rms}}| \\ &= |\mathbf{V}_{\text{rms}}| \left| \frac{\mathbf{V}_{\text{rms}}}{\mathbf{Z}} \right| \\ &= \left| \frac{\mathbf{V}_{\text{rms}}^2}{\mathbf{Z}} \right| \\ |\mathbf{V}_{\text{rms}}| &= \sqrt{S |\mathbf{Z}|} \\ &= \sqrt{(103.53 \times 10^3)(250)} \\ &= \sqrt{25882500} \\ &= 5.087 \text{ kV}\end{aligned}$$

Therefore, the value of rms voltage $\mathbf{V}_{\text{rms}}$ is $\boxed{5.087 \text{ kV}}$.

